

From General Chatbot to Specialized Assistant: ToolHive AI as a Modular, Prompt-Engineered Generative AI Platform

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I. PROJECT OVERVIEW

a. Nature of the Proposed System

ToolHive AI is a modular Generative AI (GenAI) tools library developed as a Streamlit-based web application. Grounded in the growing body of research on modular, task-oriented AI assistant architectures, the platform allows users to discover, launch, and even create their own AI tools through a unified interface, rather than relying on a single general-purpose chatbot (Oelen & Auer, 2025; Armstrong, 2025). Each tool is purpose-built for a specific task or user role, addressing the well-documented limitation of general-purpose chatbots in producing focused, reliable outputs for domain-specific workflows.

The application begins with an onboarding page that introduces the platform, explains the concept of task-specific AI assistants, and guides the user toward the main dashboard. The core of the app is the Tools Library dashboard, which presents all available tools as searchable, filterable cards. Users may launch any existing tool or create a new one through the Add New Toolkit interface, consistent with the modular, user-extensible design philosophy documented in recent AI platform research (Oelen & Auer, 2025).

The system is built on Python and Streamlit, using locally hosted large language models (LLMs) via Ollama with the Llama 3.2 model as the primary AI inference engine. Its flagship AI engineering capability is a Retrieval-Augmented Generation (RAG) pipeline powering a Policy and Data Protection Officer (DPO) Assistant. The initial toolkit includes eight specialized tools: an Interview Coach, Document Summarizer, Document Paraphraser, University Grade Predictor, Roleplay Creator, Wellness Companion, Fact Checker and Fake News Detector, and a Career Roadmap Advisor.

ToolHive AI is a capstone prototype and is not intended as a production enterprise platform. It demonstrates how different GenAI workflows can be packaged into reusable, guided tools that serve distinct user needs more effectively than a generic chatbot interface (Zhao et al., 2024; Gao et al., 2024).

b. Problem/Need Addressed

The rapid adoption of generative AI in the Philippines and globally has created both significant opportunity and a critical usability gap. As of 2024, 46% of Filipino workers utilize AI tools at least once a month, surpassing the global average of 39%, according to the JobStreet by SEEK Decoding Global Talent Report 2024. Despite this high adoption rate, experts note that the country has high adoption but low maturity or readiness in terms of AI tools, with most users relying on commercial AI services primarily for content-related tasks rather than in-production work that generates real productivity gains (Ligot as cited in PhilSTAR Life, 2026).

This gap is compounded by the nature of current AI tools available to most users. General-purpose chatbots produce unfocused, inconsistent, or poorly structured outputs when approached with highly specific professional tasks such as interview preparation, policy compliance queries, document paraphrasing, or academic grade prediction. Users are expected to craft detailed prompts themselves, which requires AI literacy that most users have not yet developed. The 2024 Microsoft and LinkedIn Work Trend Index found that Filipino employees are leading globally in leveraging AI for productivity, yet training gaps remain, particularly in using AI effectively for specific roles and functions (Microsoft & LinkedIn, 2024).

At a national level, the Department of Trade and Industry launched the National AI Strategy Roadmap (NAISR) 2.0 in July 2024, reflecting recent advancements in generative AI, though the Philippines has yet to enact legally binding AI regulations (UNESCO, 2025). This regulatory immaturity means organizations relying on AI for compliance, HR, or policy tasks face additional risk when using general-purpose tools that do not enforce structured, auditable outputs. Meanwhile, the International Labour Organization (2026) estimates that more than one-quarter of Philippine employment, approximately 12.7 million workers, is exposed to generative AI, representing the highest exposure rate among ASEAN countries with comparable data.

ToolHive AI addresses this problem by packaging different AI workflows into focused, task-specific assistants. Each tool is designed with a clear purpose, target user, guided input format, and structured output, reducing the burden on the user to prompt the AI correctly and increasing the reliability and usefulness of responses across professional, academic, and personal use cases.

c. Key Features and Functionalities

ToolHive AI is organized around three navigational layers: an onboarding page, a Tools Library dashboard, and individual tool pages. The following highlight how the system achieves its objectives:

Onboarding Page: The first screen a user encounters. It introduces ToolHive AI, explains the concept of a modular AI tools library, previews available tools, and guides the user to the main dashboard. This page establishes context and sets expectations before the user begins interacting with any tool.

Tools Library Dashboard: The main homepage of the application. Displays all available tools as browsable, filterable cards. Each card shows the tool name, target user, short description, AI capability tags, and a Launch button. A search bar and tag filter allow users to quickly locate relevant tools. An Add New Toolkit button opens the custom tool creation form, consistent with the Capbara-style user-extensible library model.

Tool Cards (Initial Toolkit):

- **Interview Coach** — Simulates mock interview sessions with role-specific questions, evaluates user answers, and provides structured feedback and improved answer suggestions. Uses multi-turn conversation memory.
- **Document Summarizer** — Accepts pasted text and returns a structured summary, key points, and action items. Designed for students, employees, and researchers.
- **Document Paraphraser** — Rewrites user-submitted text to improve clarity, change tone (formal, casual, academic), or avoid redundancy while preserving the original meaning.
- **University Grade Predictor** — Uses the LLM to help students estimate academic performance based on self-reported factors such as study hours, attendance, past grades, and assignment completion. Informed by educational data mining research (Ahmed et al., 2024; Rico-Juan et al., 2024).
- **Roleplay Creator** — Allows users to configure an AI persona such as Bill Gates, a historical leader, a job interviewer, or a fictional character, and engage in a structured roleplay conversation.
- **Wellness Companion** — A non-clinical emotional support tool that provides a judgment-free space for users to express stress, journal thoughts, or receive reflective responses. Explicitly scoped as a supportive companion, not a mental health diagnostic or therapeutic service, consistent with ethical guidelines for AI wellness tools (Moylan & Doherty, 2025; Safi et al., 2024).
- **Fact Checker and Fake News Detector** — Accepts a

news headline, claim, or article excerpt and uses the LLM with web-grounded prompting to assess credibility, identify logical inconsistencies, and suggest verification sources. Informed by NLP-based misinformation detection research (Raychaudhuri et al., 2024; Al-alshaqi et al., 2024).

- **Career Roadmap Advisor** — Helps users plan career transitions or professional development paths. Users input their current role, skills, and career goal, and the tool returns a structured roadmap with suggested learning resources, milestones, and skill gaps.

Add New Toolkit: A form-based interface where users define a custom AI tool by specifying its name, purpose, target user, system prompt, input placeholder, output format, and category. Saved tools are added to the library and immediately launchable.

d. Scope and Boundaries

The scope of ToolHive AI covers the design, development, and demonstration of a modular AI tools platform using locally hosted large language models. The system covers the full user journey from onboarding, to tool discovery on the Tools Library dashboard, to input submission, AI processing, and structured output display across eight specialized tools.

Boundaries of the system include:

- ToolHive AI is a capstone prototype and does not include production-grade security, user authentication, or multi-user account management.
- All eight tools are powered by prompt-engineered interactions with a locally hosted LLM via Ollama and do not implement retrieval-augmented generation, fine-tuning, or external database querying in this prototype version.
- The University Grade Predictor uses LLM-based inference guided by structured prompts and patterns drawn from educational data mining research. It does not train or run a standalone machine learning model on real student data.
- The Wellness Companion is explicitly not a clinical mental health tool. It does not diagnose, treat, or provide therapeutic interventions of any kind. Users experiencing mental health crises are directed to seek professional support.
- The Fact Checker tool provides LLM-assisted credibility analysis based on linguistic patterns and logical consistency of submitted text. It is not connected to a live fact-checking database and does not replace professional fact-checking organizations or verified news sources.
- The system requires a local machine with Ollama installed and the Llama 3.2 model downloaded. Internet access is not required during runtime but is needed for initial setup.
- The Custom Tool Builder saves user-created tools in session state or a local JSON file and does not include cloud storage or multi-user synchronization.

The app currently supports eight fully operational built-in tools. Additional tools may be added by users through the Add New Toolkit interface.

Intended Users and Stakeholders

The primary intended users of ToolHive AI span multiple roles and everyday contexts. **Job applicants and working professionals** benefit from the Interview Coach and Career Roadmap Advisor. **Students and researchers** benefit from the Document Summarizer, Document Paraphraser, and University Grade Predictor. **Trainers and educators** benefit from the Roleplay Creator, which enables scenario-based learning through configurable AI personas. **General users** across all backgrounds benefit from the Wellness Companion as a judgment-free space for emotional reflection, and from the Fact Checker as a tool for critically evaluating the credibility of news and online claims.

Secondary users include non-technical professionals who wish to create their own AI tools using the Add New Toolkit feature, enabling them to package their own workflows and prompts into a launchable tool without writing a single line of code. Key stakeholders include the bootcamp instructors and evaluators assessing the project as a capstone deliverable, prospective employers or development teams who may adopt or extend the platform, and the broader community of Filipino AI practitioners and learners aligned with the national AI upskilling agenda (Lumify Group, 2025; DTI, 2024).

e. Expected Benefits

ToolHive AI provides a structured, modular, and locally deployable platform that demonstrates how generative AI can be made more accessible and practically useful through task-specific design. Rather than expecting users to craft their own prompts for a generic chatbot, ToolHive AI handles the prompt engineering internally for each tool, presenting users with a guided, purpose-built interface for their specific need.

Key benefits include:

- **For Job Applicants and Professionals:** Structured, role-specific interview practice and career planning with actionable AI-generated feedback, reducing reliance on expensive coaching services.
- **For Students and Researchers:** Fast, structured summarization, tone-aware paraphrasing, and guided grade prediction support that improves academic productivity and writing quality.
- **For Educators and Trainers:** Configurable AI roleplay personas that support scenario-based learning, allowing trainers to simulate real-world conversations without requiring another person.
- **For General Users:** A judgment-free space for emotional reflection and wellness journaling through the Wellness Companion, and a practical tool for evaluating news credibility through the Fact Checker, both of which address growing concerns around mental wellbeing and misinformation in the Philippines (Safi et al., 2024; Polu, 2024).
- **For Non-Technical Users:** The Add New Toolkit feature enables anyone to create and launch a specialized AI assistant without coding knowledge, lowering the barrier to AI tool creation significantly.

For the Philippine AI Ecosystem: ToolHive AI demonstrates a practical, low-cost, privacy-preserving GenAI application

that runs entirely on local hardware using open-source models, making it accessible even in resource-constrained settings and aligned with the goals of the National AI Strategy Roadmap 2.0 (DTI, 2024).

II. CORE DATA REQUIREMENTS

a. Major System Modules and Workflows

1. Onboarding Module

The first screen rendered when the application loads. Introduces ToolHive AI, explains the task-specific assistant concept, previews available tools, and routes the user to the Tools Library dashboard.

- Display platform name, tagline, and short description
- Show preview cards or icons of available tools
- Provide Get Started button routing to the dashboard

2. Tools Library Dashboard Module

The main navigation hub of the application. Renders all available tools as cards from a combined registry of built-in and user-created tools.

- Load built-in tool definitions from internal configuration
- Load custom tools from custom_tools.json
- Render filterable, searchable tool cards
- Display Add New Toolkit button routing to the tool builder form
- Route user to the selected tool page on card launch

3. Interview Coach Module

A multi-turn conversational AI tool for interview practice. Maintains session history in Streamlit session state.

- Accept role or domain selection from user
- Generate role-specific interview questions
- Evaluate user answers and return structured feedback
- Suggest improved responses

4. Document Summarizer Module

A single-turn structured output tool for text reduction and extraction.

- Accept long-form text input
- Return summary, key points, and action items in a formatted layout

5. Document Paraphraser Module

A tone-aware text rewriting tool.

- Accept text input and tone selection (formal, casual, academic, simplified)
 - Return rewritten version preserving original meaning
- Display original and paraphrased text side by side

6. University Grade Predictor Module

An LLM-guided academic performance estimation tool. Accepts structured self-reported input and returns a predicted performance estimate with explanations and improvement suggestions, aligned with educational data mining research (Ahmed et al., 2024).

- Accept inputs: study hours per week, attendance rate, past grades, assignment completion rate, self-assessed engagement
- Generate structured performance estimate and recommendation

7. Roleplay Creator Module

A persona configuration and conversation tool for scenario-based interaction.

- Accept persona configuration: name, role, background, behavioral tone
- Build system prompt from configuration
- Maintain multi-turn conversation memory using session state
- Generate in-character responses

8. Wellness Companion Module

A non-clinical emotional support and reflective journaling assistant. Designed according to guidance that AI wellness tools should explicitly communicate their non-clinical limitations and avoid simulating therapeutic diagnoses (Moylan & Doherty, 2025).

- Display a clear non-clinical disclaimer on entry
- Accept open-ended emotional or reflective input
- Return empathetic, supportive, non-prescriptive responses
- Optionally prompt the user with journaling questions

9. Fact Checker and Fake News Detector Module

An LLM-assisted credibility analysis tool. Informed by NLP-based fake news detection methodologies (Raychaudhuri et al., 2024; Polu, 2024).

- Accept news headline, claim, or article excerpt
- Analyze linguistic patterns, logical consistency, and claim structure
- Return a credibility assessment with reasoning
- Suggest verification sources for the user to consult

10. Career Roadmap Advisor Module

A structured career planning tool.

- Accept current role, existing skills, and target career goal
- Return a structured roadmap with skill gaps, learning milestones, and recommended resources

11. Add New Toolkit (Custom Tool Builder) Module

A form-based tool creation interface for user-defined tools.

- Accept tool name, description, target user, category, system prompt, input placeholder, output format
- Save tool to custom_tools.json

Display newly created tool as a launchable card in the Tools Library

a. Key Data Requirements

COMPONENT	PURPOSE	IMPLEMENTATION
PYTHON 3.10+	Core programming language	Required runtime
STREAMLIT	Multi-page web application framework	Pages folder + session state
OLLAMA	Local LLM inference engine	REST API at localhost:11434
LLAMA 3.2	Primary language model	Pulled via Ollama CLI
SENTENCE TRANSFORMERS	Semantic similarity for RAG	Used in retrieval.py
COSINE SIMILARITY / VECTOR SEARCH	Top-k chunk retrieval	NumPy or FAISS
JSON / TEXT FILES	Document and tool storage	policy_docs/ and custom_tools.json
STREAMLIT SESSION STATE	Conversation memory	st.session_state across pages

III. SYSTEM ARCHITECTURE AND AI WORKFLOW

ToolHive AI is designed using a modular multi-page Streamlit architecture centered on a structured three-layer workflow: Onboarding Page → Tools Library Dashboard → Individual Tool Pages. This architectural approach separates user onboarding, tool discovery, and AI execution into independent layers, improving both usability and maintainability. Such modular separation is consistent with modern AI platform design principles discussed in recent literature (Oelen & Auer, 2025).

The application entry point is handled through app.py, which renders the onboarding interface. The Tools Library dashboard (pages/0_Tools_Library.py) functions as the central navigation hub where users can browse and access available tools. Each AI tool is implemented as a dedicated Streamlit page inside the pages/ directory, allowing every module to maintain isolated logic and independent functionality.

Shared services such as the Ollama API client and reusable interface components are centralized in the utils/ directory to improve code reuse and maintainability. User-generated custom tools are stored in data/custom_tools.json and dynamically loaded into the dashboard during runtime.

Recommended folder structure:

```
toolhive_ai/  
  app.py  
  pages/  
    0_Tools_Library.py  
    1_Interview_Coach.py = iris  
    2_Document_Summarizer.py = iris  
    3_Document_Paraphraser.py = iris  
    4_Grade_Predictor.py = iris  
    5_Roleplay_Creator.py = lloyd  
    6_Wellness_Companion.py = lloyd  
    7_Fact_Checker.py = lloyd  
    8_Career_Roadmap.py = lloyd  
    9_Custom_Tool_Runner.py  
  frontend/  
    textStyle  
  data/  
    custom_tools.json  
  utils/  
    ollama_client.py  
    ui.py  
  requirements.txt  
  README.md
```

b. Three-Layer User Journey

Layer 1 — Onboarding Page

The onboarding page serves as the initial interface displayed when ToolHive AI is launched. Its primary purpose is to introduce users to the platform, explain the concept of a modular AI-powered tools library, and provide an overview of the available functionalities. The page also includes a clear call-to-action directing users to the Tools Library dashboard. This layer helps reduce the learning curve by establishing user expectations before interaction with any AI-powered feature begins.

Layer 2 — Tools Library Dashboard

The Tools Library dashboard acts as the primary navigation and discovery interface of the application. It displays all available tools as searchable and filterable cards generated from both built-in tool definitions and user-created tools stored in custom_tools.json.

Each tool card contains the following information:

- Tool name
- Target user
- Short description
- AI capability tags
- Launch button

To improve accessibility and usability, the dashboard includes a search bar and category filters that allow users to quickly locate relevant tools. An “Add New Toolkit” button opens a custom tool creation form where users can define and save their own AI-powered tools without modifying source code. This centralized tools library concept aligns with modular AI platform interfaces described in recent research

(Armstrong, 2025).

Layer 3 — Individual Tool Pages

Each tool is implemented as a dedicated Streamlit module with its own interface, prompt logic, and output layout. Instead of relying on generic prompting, every tool is designed for a specific purpose and only exposes the inputs necessary for that task. Prompt engineering is handled internally by the application, allowing users to interact with the tools without needing technical knowledge of prompt construction.

c. AI Inference Architecture

All tools in ToolHive AI share a unified AI inference architecture powered by a locally hosted large language model served through Ollama. Communication with the model occurs through Ollama’s REST API running at localhost:11434, while the primary language model used throughout the system is Llama 3.2.

Using a locally deployed model provides several advantages for a capstone-scale prototype. First, it removes dependency on paid cloud APIs, eliminating operational costs. Second, all processing occurs on the user’s machine, improving data privacy and minimizing external data exposure. Third, the platform remains functional even without internet connectivity during runtime (Jabarian, 2024; Belsare, 2024).

Each tool interacts with the shared Ollama client utility located in utils/ollama_client.py. The tool module constructs a prompt using user inputs and predefined system instructions, sends the request to the Ollama API, and displays the generated response using an output format tailored to the tool’s intended functionality.

d. Prompt Engineering Strategy

Since ToolHive AI does not implement retrieval-augmented generation (RAG) or model fine-tuning, prompt engineering serves as the primary AI customization mechanism. Each tool uses a carefully structured system prompt that defines the AI’s role, behavior, response format, tone, and operational boundaries for its intended task. Prior research has shown that task-specific prompting significantly improves output reliability and usefulness compared to unrestricted prompting approaches (Zhao et al., 2024; Gao et al., 2024).

The prompt engineering strategies for each tool are summarized below:

Interview Coach

The AI is configured to act as a professional interviewer tailored to a user-selected role or industry. It asks one interview question at a time, evaluates the user’s response based on clarity, relevance, and depth, and provides suggestions for improvement before proceeding to the next question.

Document Summarizer

The system prompt instructs the model to generate a structured output containing:

1. A concise summary
2. Key points in bullet format
3. Action items when applicable

The prompt emphasizes brevity and discourages unnecessary commentary.

Document Paraphraser

The AI rewrites user-provided text according to a selected tone such as formal, casual, academic, or simplified. The system prompt ensures that the original meaning is preserved while modifying vocabulary, sentence structure, and writing style.

University Grade Predictor

The model acts as an academic performance advisor that analyzes inputs such as study hours, attendance, previous grades, assignment completion, and self-assessed engagement. It generates a predicted performance estimate together with explanations and personalized recommendations. The prompt structure is based on factors identified in educational data mining studies (Ahmed et al., 2024; Rico-Juan et al., 2024).

Roleplay Creator

The AI persona is dynamically constructed using user-defined settings such as character name, role, background, and behavioral tone. The prompt instructs the model to remain fully in character throughout the interaction unless explicitly instructed otherwise.

Wellness Companion

The system prompt defines the AI as a non-clinical emotional support companion. The AI is explicitly restricted from diagnosing medical conditions or simulating professional therapy. Instead, it responds empathetically, asks reflective questions, and encourages professional support-seeking when serious distress is detected. This approach aligns with ethical guidelines for AI wellness systems (Moylan & Doherty, 2025; Safi et al., 2024).

Fact Checker and Fake News Detector

The AI evaluates submitted content based on:

- Linguistic credibility indicators
- Logical consistency
- Alignment with general knowledge

The tool produces a structured credibility assessment and recommends source categories users should consult for verification (Raychaudhuri et al., 2024; Polu, 2024).

Career Roadmap Advisor

The AI functions as a professional development assistant that analyzes the user's current skills, present role, and target career objective. It then generates a structured roadmap containing skill gaps, recommended milestones, and prioritized learning resources.

e. Conversation Memory Architecture

Three tools within ToolHive AI support multi-turn conversations: Interview Coach, Roleplay Creator, and Wellness Companion. These tools utilize Streamlit session

state (`st.session_state`) to preserve conversation history during runtime sessions.

Whenever the user submits a new message, the complete message history is passed to the Ollama API as a structured messages array. This enables the language model to maintain contextual continuity and generate responses that reference earlier parts of the interaction.

The remaining tools — Document Summarizer, Document Paraphraser, University Grade Predictor, Fact Checker, and Career Roadmap Advisor — operate as single-turn systems. Each request is processed independently without retaining prior interactions, which is appropriate for their task-oriented workflows.

f. Custom Tool Execution Architecture

ToolHive AI supports user-generated AI tools through the Add New Toolkit feature. Custom tools are stored as structured JSON objects inside `data/custom_tools.json`.

Each custom tool configuration contains:

- Tool name
- Description
- Target user
- Category tag
- System prompt
- Input placeholder text
- Output format instruction

When a user launches a custom tool from the dashboard, the generic execution interface (`pages/9_Custom_Tool_Runner.py`) loads the saved configuration, initializes the session using the stored system prompt, and processes interactions through the same Ollama client utility used by the built-in tools.

This architecture enables users to create and execute an unlimited number of custom AI-powered tools without writing or modifying application code, significantly improving extensibility and scalability within the platform.

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